

SESSION 4

RAD-423 Imaging Informatics

Cloud Computing & Cloud-Based Medical Imaging

Cloud PACS, service models, secure access, resilience and radiology workflow

Course	RAD-423 Imaging Informatics
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Lecturer	A. A. Zubair
Suggested study time	60–90 minutes

Learning Objectives

By the end of this session, the student should be able to:

1. define cloud computing and explain what cloud-based imaging means in radiology;
2. compare traditional on-premises PACS with cloud-based PACS;
3. explain the major advantages of cloud imaging, including scalability, remote access, collaboration and disaster recovery;
4. discuss important limitations involving connectivity, privacy, security, cost and vendor lock-in;
5. distinguish Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS); and
6. describe the practical responsibilities of imaging professionals when clinical workflows depend on cloud services.

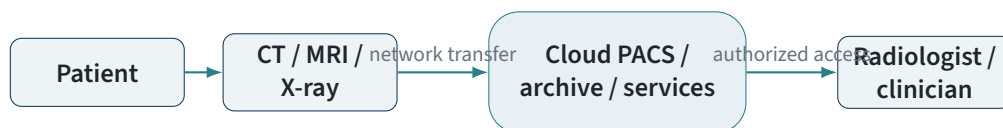
Key Concept

Cloud imaging changes where computing and storage resources are hosted; it does not remove the need for good radiology workflow, secure access, reliable networks or local contingency planning. The modality still acquires the examination, but storage, viewing, reporting or other services may be delivered from remote cloud infrastructure.

1 What Is Cloud Computing?

Cloud computing is the use of computing resources—including storage, servers, databases, applications and processing power—over a network, usually the internet. In medical imaging, cloud infrastructure can host images, archives and applications that authorized users access without all of the underlying hardware being located inside the hospital.

A useful way to think about cloud imaging is to separate the **clinical activity** from the **location of the computing resource**. The patient is still scanned in the imaging department, but some of the systems that store, process or display the resulting data may be hosted elsewhere.



Common Error

Do not equate **cloud imaging** with simply uploading a folder of JPEG images to an online drive. A clinical cloud-imaging environment must preserve the imaging workflow, patient/study information, access controls, retrieval performance and other requirements needed for safe clinical use.

1.1 What may be hosted in the cloud?

Depending on the design, cloud infrastructure may support one or more of the following:

- image storage and long-term archiving;
- PACS viewing and distribution;
- remote reporting and teleradiology;
- disaster-recovery copies;
- image exchange between approved institutions;
- analytics or artificial-intelligence workloads; and
- selected applications that would otherwise run on local servers.

Not every component must move to the cloud. Many healthcare organizations use a **mixed architecture**: modalities and some workflow components remain local while archives, viewers or other services are remotely hosted.

2 Traditional PACS versus Cloud PACS

A useful way to understand cloud PACS is to compare it with a traditional on-premises PACS.

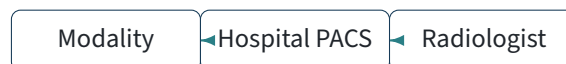
Feature	Traditional / on-premises PACS	Cloud-based PACS
Core infrastructure	Hospital or local data-centre servers	Provider-hosted or remotely managed infrastructure
Storage growth	Requires local capacity planning and hardware expansion	Can be expanded through contracted cloud resources
Access	Usually optimized for institutional network access	Can support secure access from multiple approved locations
Local hardware burden	Greater responsibility for local servers and storage	Some infrastructure responsibility shifts to provider/managed service
Connectivity dependence	Internal network remains important	External connectivity becomes especially important
Cost pattern	Often larger capital infrastructure purchases	Often recurring operational/service costs
Management	Institution manages more of the technical stack	Responsibilities are divided between institution and provider

High-Yield

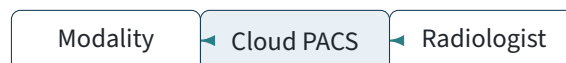
The essential difference is not that one system uses a network and the other does not. Both do. The major difference is where key computing/storage resources are hosted and who is responsible for operating them.

2.1 A simplified workflow comparison

Traditional



Cloud-based



Cloud-based access can make remote viewing and reporting easier, but the reliability and security of the network path then become part of the clinical workflow itself.

3 Advantages of Cloud-Based Imaging

Major advantages of cloud-based imaging include **scalability, remote access, disaster recovery, reduced local infrastructure and collaboration.**

3.1 1. Scalability

Medical imaging generates large and continually increasing data volumes. CT, MRI, digital radiography, ultrasound cine loops and advanced reconstructions can consume substantial storage. Cloud services can allow an organization

to increase contracted storage or processing capacity without installing equivalent amounts of new local hardware each time demand grows.

Example: growing image volume

A department introduces a new CT scanner and begins performing more thin-slice examinations. The number of images per study rises sharply. In a fully local architecture, the organization may need to purchase and install additional storage. In a cloud-based architecture, capacity may be increased through the service arrangement, although the financial and contractual consequences still need to be considered.

3.2 2. Remote access and teleradiology

Authorized users can potentially access studies from approved locations outside the physical imaging department. This supports:

- teleradiology;
- remote reporting;
- on-call interpretation;
- multidisciplinary review; and
- multicentre clinical or research collaboration, where governance permits.

Remote access does not mean unrestricted access. Authentication, authorization, secure transmission and organizational policy remain essential.

3.3 3. Disaster recovery

Geographically distributed cloud infrastructure can help maintain copies of important information away from the primary hospital site. This may improve resilience against local hardware failure, fire, flood or other site-specific disruption.

Backup is not the same as availability

A remote copy is useful only if the organization knows how services will actually be restored and accessed during an outage. Disaster recovery therefore requires more than storing a second copy; it requires tested procedures, defined responsibilities and an operational recovery pathway.

3.4 4. Reduced local infrastructure

A cloud-based model can reduce the number or scale of servers that an institution must purchase, house, cool, power and replace locally. It does **not** eliminate local technical responsibility. The institution still needs networking, endpoint security, account management, governance and support for the clinical workflow.

3.5 5. Collaboration

Cloud platforms can make it easier to share studies among approved sites or teams because users do not necessarily need to be physically connected to the same local PACS. This can support referral pathways, specialist consultation and distributed reporting when the necessary privacy and security arrangements are in place.

4 Challenges and Limitations

Cloud-based imaging introduces a different set of operational risks. These risks do not make cloud imaging unsuitable, but they must be understood, planned for and managed appropriately.

4.1 Internet and network dependence

If the imaging department depends on remote services, a network failure can interrupt image upload, retrieval or reporting. This is particularly important where connectivity is slow, unstable or expensive.

Two concepts matter:

Bandwidth

How much data can be transferred in a given time. Large CT or MRI studies may require substantial bandwidth.

Latency

Delay in communication between systems. High latency can make interactive image viewing feel slow even when a connection is technically available.

Clinical / Practical Link

A hospital may have a functioning CT scanner but temporarily lose its external connection to the cloud PACS. Acquisition may still be possible, but normal transfer, retrieval or reporting may be disrupted. A safe design therefore needs a documented downtime workflow rather than assuming continuous connectivity.

4.2 Security and privacy

Patient imaging information remains sensitive regardless of where it is stored. Cloud use therefore requires appropriate controls for identity, access, encryption, logging, monitoring, data handling and incident response. Responsibilities may be shared with a service provider, but the healthcare organization cannot simply transfer all accountability to that provider.

4.3 Cost

Cloud services can reduce some capital expenditure but introduce recurring operational costs. Cost may depend on factors such as:

- total storage volume;
- computing resources used;
- number of users or applications;
- data transfer and retrieval;
- backup and retention requirements; and
- service/support agreements.

A system that appears inexpensive at initial deployment may become costly if data volume and retrieval requirements grow substantially.

4.4 Vendor lock-in

Vendor lock-in occurs when moving applications or large datasets away from one provider becomes difficult, technically complex or expensive. In imaging this matters because archives may contain years of studies. Procurement decisions should therefore consider data portability, standards support, migration procedures and exit arrangements.

Common Error

Do not judge cloud imaging only by the monthly storage price. The technically important question is the **whole service**: storage, retrieval, performance, data transfer, security, backup, support, migration and downtime arrangements.

5 Cloud Service Models: IaaS, PaaS and SaaS

Three common cloud service models are IaaS, PaaS and SaaS. They differ mainly in **how much of the technical stack is managed by the provider versus the customer**.

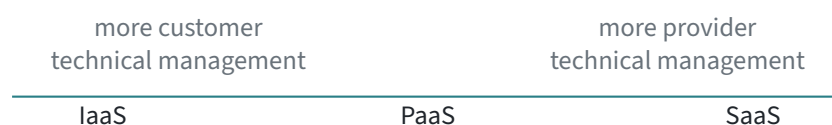
Model	Name	Provider mainly supplies	Imaging-oriented example
IaaS	Infrastructure as a Service	Computing, storage and networking infrastructure	An organization rents cloud servers/storage and installs or manages its imaging applications on them
PaaS	Platform as a Service	A managed platform for building or running applications	A development/analytics platform used to build imaging workflow or data applications
SaaS	Software as a Service	A complete application delivered as a service	A web-accessible cloud PACS/viewer where the user does not manage the underlying servers

Memory aid

IaaS = rent the infrastructure. PaaS = use a managed platform. SaaS = use the finished software service.

5.1 Responsibility changes across the models

As the service moves from IaaS toward SaaS, the provider manages more of the underlying technical environment. However, the healthcare organization still retains important responsibilities such as user authorization, correct clinical use, patient-data governance and local workflow management.



6 Cloud Deployment Models

Cloud deployment models describe **who the infrastructure is intended for and how different environments are combined**. Common models include **private, community, public and hybrid** cloud. These terms are different from IaaS, PaaS and SaaS, which describe the level of service provided rather than the ownership or arrangement of the cloud environment.

Private cloud

Cloud infrastructure used exclusively by one organization or defined organizational group.

Community cloud

Infrastructure intended for a specific community of organizations with shared concerns or requirements.

Public cloud

Infrastructure offered for use by the general public or a broad group of customers by a provider.

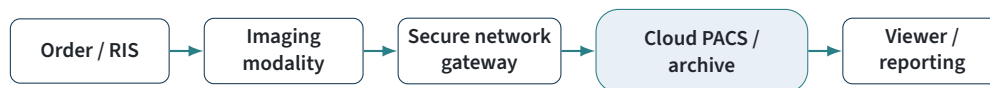
Hybrid cloud

A combination of distinct cloud environments linked so that data or applications can move between them according to the design.

In healthcare, a hybrid approach may be particularly practical: some services remain within the institution while selected archives, applications, backup copies or computational workloads use externally hosted infrastructure.

7 Cloud-Enabled Radiology Workflow

A cloud-enabled workflow can be represented as follows:



At each step in the workflow, four questions should be considered:

1. **What information is being transferred?**
2. **Who is authorized to access it?**
3. **What happens if the connection or service is unavailable?**
4. **How will the organization verify that the correct study reached the correct destination?**

7.1 Downtime and contingency planning

Cloud-dependent services need a plan for periods when the remote service or network cannot be reached. Depending on local architecture, this might include temporary local storage, alternative viewing arrangements, communication procedures and later reconciliation of studies once connectivity returns.

Scenario: temporary loss of cloud connectivity

An MRI examination is completed successfully, but the department cannot reach the cloud PACS. The correct response is not simply to repeat the examination. The team should follow the approved downtime process: preserve the acquired study locally if the system supports it, notify the appropriate technical/clinical teams, use the approved alternative reporting pathway where available, and confirm later that the study has

been transmitted and reconciled correctly.

8 Role of the Radiographer

Radiographers do not normally design cloud infrastructure, but their daily actions influence whether the cloud workflow is safe and reliable. Important responsibilities include:

- ensuring correct patient and examination identification before transmission;
- confirming that studies have been sent successfully to the intended destination;
- recognizing failed or delayed transfers;
- protecting login credentials and using only approved access methods;
- following policy for remote access and patient-data handling;
- understanding local downtime procedures; and
- reporting unusual access, transfer or availability problems promptly.

Exam focus

A good answer about cloud imaging should be balanced. Do not write only “cloud is cheaper and accessible anywhere.” Discuss both the benefits and the trade-offs: scalability and remote access must be considered alongside connectivity, privacy, security, recurring cost, service availability and vendor dependence.

Key Takeaways

- Cloud computing provides remote access to configurable computing resources such as storage, servers, applications and processing power.
- Cloud-based imaging may host PACS, archives, viewers, reporting tools, backup copies or other imaging services.
- Traditional PACS is mainly hosted locally; cloud PACS shifts more infrastructure to a remote provider or managed service.
- Major benefits include scalability, remote access, collaboration, disaster-recovery options and reduced local infrastructure burden.
- Major challenges include network dependence, security, privacy, ongoing cost and vendor lock-in.
- IaaS, PaaS and SaaS describe different service models and different divisions of technical responsibility.
- Private, community, public and hybrid describe deployment models and are distinct from IaaS/PaaS/SaaS.
- A cloud strategy still requires downtime procedures, secure access and careful clinical workflow management.

Key Terms

Cloud computing

Network-based access to configurable computing resources such as servers, storage and applications.

Cloud PACS

A PACS in which important application or storage components are hosted using cloud infrastructure.

Scalability

Ability to increase or decrease computing/storage capacity as demand changes.

Teleradiology

Remote interpretation of imaging examinations using electronic communication systems.

Latency

Communication delay between systems.

Bandwidth

Amount of data that can be transferred over a connection in a given time.

Vendor lock-in

Difficulty or cost associated with moving data or services away from a provider.

IaaS

Infrastructure as a Service.

PaaS

Platform as a Service.

SaaS

Software as a Service.

Hybrid cloud

Use of linked distinct cloud environments as part of one architecture.

Self-Assessment Questions

1. Define cloud computing in the context of medical imaging.
2. Compare a traditional on-premises PACS with a cloud-based PACS.
3. Explain three advantages of cloud-based medical imaging.
4. Why can poor connectivity become a clinical workflow problem in a cloud-PACS environment?
5. Differentiate bandwidth from latency.
6. Explain vendor lock-in and why it matters for long-term imaging archives.
7. Distinguish IaaS, PaaS and SaaS and give an imaging-related example of each.
8. What is a hybrid cloud, and why might a healthcare organization use a hybrid approach?
9. An MRI study cannot be sent because the cloud service is temporarily unavailable. Explain the principles that should guide the radiographer's response.

